



Pascal was a 17th century French philosopher, theologian, and mathematician; he made foundational contributions to, among other areas, the early development of the theory of probability.

Pascal was one of the first thinkers to systematically investigate the question of how we should make decisions under situations of uncertainty, where we don't know all of the relevant facts about the world, or the outcomes of our actions. He was (with his contemporary Fermat) the first to formulate the idea of **expected utility**.

Let's look at one last idea about basic beliefs. This is the idea that we should sometimes have a belief for **practical** reasons. Our focus will be a famous argument from Blaise Pascal which makes use of this idea.

The expected utility of an action can be calculated by looking at the various possible outcomes of the action, and assigning each a value — measuring how good the outcome is — and a probability — measuring how likely the outcome is. Because you know that one of the outcomes is going to happen (but not more than one) the probabilities should sum to 1. To get the expected utility, you multiply each outcome's value by its probability, and add them all up.

So consider a bet in which a fair coin is flipped. Suppose that you get \$5 if it comes up heads, and lose \$3 if it comes up tails. Then the expected utility is:

$$(0.5 * \$5) + (0.5 * \$3) = \$1$$

Many have thought that expected utility considerations should guide our actions. Perhaps we should all act so as to maximize the expected utility of our actions. (We'll come back to this idea in a few weeks.)

Pascal had the thought that forming beliefs is just one sort of action we perform. So, if in general it makes sense for our actions to be guided by expected utility, why not also form beliefs on the basis of expected utility?

This suggests the following rule of belief:

Expected Utility $\rightarrow$ Belief

If believing P has a higher expected utility than not believing P , you should believe P .

This rule of belief led Pascal to a famous argument for belief in God.

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